

Data Analyst Technical Assessment

Business Context

You're working with a DTC (Direct-to-Consumer) fitness supplements brand that has been running multichannel marketing campaigns across Facebook, Google, and TikTok. The company currently:

- Spends \$1.6M monthly across all channels
- Targets a minimum ROAS of 1.2 (industry benchmark: 1.4)
- Aims to grow revenue by 40% next quarter while maintaining profitability
- Operates in a highly seasonal market with key competitors running major promotions during holiday periods
- Currently underperforming with an overall ROAS of 0.86, well below the target of 1.2

Challenge

The marketing team needs to optimize their current spend allocation and strategy to improve performance. They face several critical challenges:

- Current ROAS is significantly below target (0.86 vs 1.2)
- Dramatic performance variation across platforms, creatives, and product categories
- Declining efficiency in recent weeks (weekly ROAS trending downward)
- Unclear regional performance patterns
- Uncertain impact of competitive events
- Limited understanding of creative performance and audience targeting effectiveness
- Need to cut spend by 30% in underperforming areas while growing revenue

Dataset

The provided dataset ( Marketing Data) contains 3 months of daily marketing performance data (January-March 2024) with the following structure:

- Date
- Platform (Facebook, Google, TikTok)
- Campaign Name
- Region (West, South, Northeast, Midwest)
- Ad Spend
- CPM
- Impressions

- Frequency
- Clicks
- Purchases
- Revenue
- Product Category (Pre-workout, Protein, Weight Loss, Diet)
- Target Audience (Athletes, Fitness Enthusiasts, Weight Loss)
- Creative Type (Video, Image, Carousel, Search, Display)
- Video Completion Rate
- Customer LTV
- Is_Competitive_Event flag

Note: The dataset contains some reporting inconsistencies and missing values that will need to be addressed.

Task Components

1. Data Quality & Exploration

Examine and prepare the dataset for analysis. Your submission should include:

Primary Deliverable:

- A data quality assessment summary that documents your data cleaning process and approach
- Clear identification of any data issues and how you addressed them

Key Areas to Address:

- Data completeness and reliability (identify and handle missing values)
- Performance metric validation and calculation (CTR, CPC, CVR, ROAS)
- Unusual patterns or outliers (identify and address extreme values)
- Key assumptions that could impact recommendations

Guiding Questions:

- What data quality issues did you find and how did you address them?
- Which performance metrics show the most concerning trends?
- What baseline metrics are most relevant for this business given the challenges?
- What preliminary insights can you derive from the data before deeper analysis?

2. Performance Analysis

Conduct a comprehensive analysis to identify performance patterns and opportunities:

Required Analyses:

- Channel performance analysis with statistical significance testing
 - Compare platforms (Facebook, Google, TikTok) across key metrics
 - Identify which platform is driving the most value and why
- Regional performance comparison
 - Analyze regional differences with statistical significance testing
 - Identify over/underperforming regions and potential reasons
- Creative performance analysis
 - Compare creative types (Video, Image, Carousel, Search, Display)
 - Analyze the relationship between video completion rate and conversion
- Product category and audience targeting performance
 - Identify which products and audience segments deliver the best ROAS
- Week-over-week trend analysis to identify performance patterns
- Impact of competitive events on key metrics
- Relationship between ad frequency and conversion rates

Deliverable:

- Technical analysis document with:
 - Clearly documented methodology for each analysis
 - Statistical tests and significance levels used
 - Visualizations highlighting key findings
 - Interpretations of findings in business context

3. Strategic Recommendations

Develop an actionable marketing optimization plan addressing:

a) Immediate Optimization Opportunities

- Develop a clear framework for cutting 30% of spend from the lowest-performing areas
- Specifically identify which campaigns, platforms, regions, or creative types should be reduced or eliminated
- Recommend which high-performing channels/campaigns should receive additional budget
- Propose specific bidding strategy adjustments around competitive events based on performance data

b) Testing Recommendations

- Propose 3 specific A/B tests to improve performance
- Include for each test:
 - Clear hypothesis based on data findings
 - Test methodology and experimental design
 - Success metrics and measurement approach
 - Expected impact with quantitative estimates

- Timeline and implementation plan

c) Long-term Strategy

- Develop a data-driven regional budget allocation strategy
- Propose a creative strategy by platform based on performance data
- Recommend audience and product category targeting priorities
- Outline a quarterly planning approach with competitive event adjustments

Deliverable:

- Executive summary (max 750 words) including:
 - 3-5 key insights from your analysis
 - Prioritized recommendations with expected impact estimates
 - Implementation timeline with specific actions
 - Risk assessment and mitigation strategies
 - Measurement plan to track implementation success

4. Advanced Analytics Bonus (Optional)

For candidates with advanced analytics skills, choose ONE of the following:

a) Predictive Modeling

- Build a predictive model to forecast expected ROAS by campaign
- Use historical data to identify factors that best predict performance
- Propose how this model could be used for budget allocation

b) Marketing Mix Modeling

- Develop a simple MMM to understand contribution of each channel
- Account for factors like seasonality and competitive events
- Estimate incremental return by channel

c) Customer Segment Analysis

- Use clustering techniques to identify valuable customer segments
- Propose targeted strategies for each segment
- Estimate potential revenue impact of segmentation

APPENDIX A: Glossary

Basic Performance Metrics:

- DTC (Direct-to-Consumer): Business model selling products directly to customers without intermediaries

- ROAS (Return on Ad Spend): Revenue generated per dollar of advertising spend
- CTR (Click-Through Rate): Percentage of impressions that result in clicks
- CPC (Cost Per Click): Average cost paid for each click
- CVR (Conversion Rate): Percentage of clicks that result in purchases
- CPM (Cost Per Mille): Cost per thousand impressions
- LTV (Lifetime Value): Total revenue expected from a customer over time

Ad Platform Terms:

- Campaign: Organized group of ads with shared objective
- Creative Type:
 - Carousel: Multiple scrollable images/videos in one ad
 - Display: Visual banner ads on websites
 - Search: Text ads on search engine results
 - Video: Motion-based advertising content
 - Image: Static visual advertisements

Campaign Metrics:

- Impressions: Number of times an ad is shown
- Frequency: Average number of times a user sees an ad
- Video Completion Rate: Percentage of video views watched to completion
- Competitive Event: Period with increased competitor advertising (e.g., holidays)

Marketing Strategy Terms:

- Channel: Advertising platform (e.g., Facebook, Google, TikTok)
- Creative Performance: How well different ad formats perform
- Bidding Strategy: How much to pay for ad placement
- Budget Allocation: Distribution of spending across channels/regions

Industry Context:

- Seasonal Market: Demand fluctuates based on time of year
- Industry Benchmark: Standard performance level in the industry
- Target Audience: Specific customer segments being targeted